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$$\lim_{x \rightarrow 1} \frac{x-1}{9x^2-5x-4}$$

$$f(x) = x-1 \Rightarrow f'(x) = 1$$

$$g(x) = 9x^2-5x-4 \Rightarrow g'(x) = 18x-5$$

$$\lim_{x \rightarrow 1} \frac{x-1}{9x^2-5x-4} = \lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 1} \frac{1}{18x-5}$$

$$\lim_{x \rightarrow 1} \frac{1}{18x-5} = \frac{1}{18 \cdot 1 - 5} = \frac{1}{13}$$

References